

1a	The rate of change of momentum of a body is proportional to the resultant force acting on the body and takes place in the direction of the resultant force.	B1 B1
1bi	The acceleration of object X is dependent on the resultant of the two forces, weight and tension, acting on object X. Since the <u>weight and tension are in opposite directions</u> , the <u>resultant force must be less than its weight</u> and hence, has an acceleration less than g .	B1
1bii	Let T be the tension in the rope. Considering the FBD of object X, by N2L, $\downarrow +:$ $m_X g - T = m_X a$ $T = m_X g - m_X a \dots\dots\dots (1)$ Considering the FBD of object Y, by N2L, $\uparrow +:$ $T - m_Y g = m_Y a$ $T = m_Y g + m_Y a \dots\dots\dots (2)$ Equating (1) and (2), we have $m_X g - m_X a = m_Y g + m_Y a$ $g (m_X - m_Y) = a (m_X + m_Y)$ Hence, $a = \frac{m_X - m_Y}{m_X + m_Y} g$	M1 M1 A1
1biii	Since the motion is linear with uniform acceleration, using $s = ut + \frac{1}{2} at^2$ $\downarrow +:$ $5.0 = 0 + \frac{1}{2} \left[\frac{6.0 - 3.0}{6.0 + 3.0} (9.81) \right] t^2$ $t = 1.749 \text{ s} = 1.7 \text{ s}$	C1 A1
2a	The object is in equilibrium and hence resultant force is zero	